Finding the students who will leave, while there is still time to help

An early warning on day one. And one decision only the school can make.

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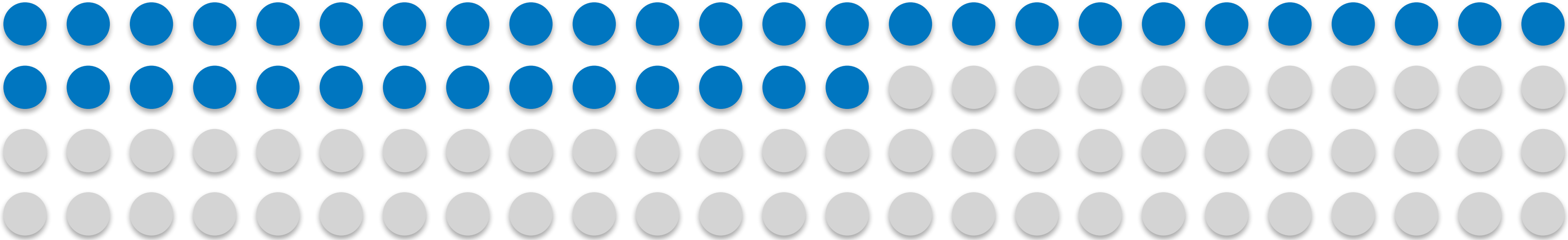
PGD AI/ML capstone, Pillar 5
Built on 4,424 student records from Instituto Politecnico de Portalegre

391 students in every 1,000 leave before they finish



One dot is ten students. The filled ones do not graduate.

Every one of them costs three things. The student loses the years and the fees. The school loses the tuition. The country loses a graduate it had already half paid for.

The school already knows this is happening. What it does not know is which students, and it does not find out until the first exam results come back, by which point half the year is gone and the student is usually already gone with it.

2,100 EUR

is the tuition a single retained student pays back over a three year degree. A deliberately low estimate. The real loss to that student, and to the country, is larger and harder to defend, so it is left out.

The question is not whether to help. It is who to help first, and when.

What the tool does, on the day a student enrolls

It reads the enrollment form. Nothing else.

Age. Course. How they applied. Whether they hold a scholarship. Whether the fees are paid. What their parents do. Everything on that list is known before the student sits in a single class.

It does not look at grades, attendance, or coursework.

That was a deliberate choice, and it costs accuracy. A tool that waits for first-semester results would be far more certain. It would also be useless, because by then the student has already decided.

It hands the tutoring team a ranked list.

Most at risk at the top. Not a label, not a verdict, and never a number shown to a student. A starting point for a conversation, twelve months before it would otherwise happen.

12 months
of warning, instead of none

8 in 10
of the students who go on to leave are on the list

1 in 2
students are flagged, which is what the tutoring team can actually handle

The tool does not decide anything. It decides what a person looks at first.

Out of every 1,000 students

Scaled from 726 real students the tool had never seen. It was tested once, on data held back from the start.

Of the 391 who leave, the tool finds 311 of them.

Eighty are missed. That is the honest number and it is not going to be zero. No tool that runs on an enrollment form is going to find everyone, and a tool that claimed to would be lying.

The 172 is the number worth arguing about, and slide 6 argues about it.

483

students on the list. That is roughly one in two, and it is set by what the tutoring team can staff, not by the tool.

Every number on this slide came from students the tool had never seen.

The return

Three assumptions, all stated, all conservative. Change any one and the answer moves, which is why the curve is here and not just the number.

A retained student is worth about 2,100 EUR.

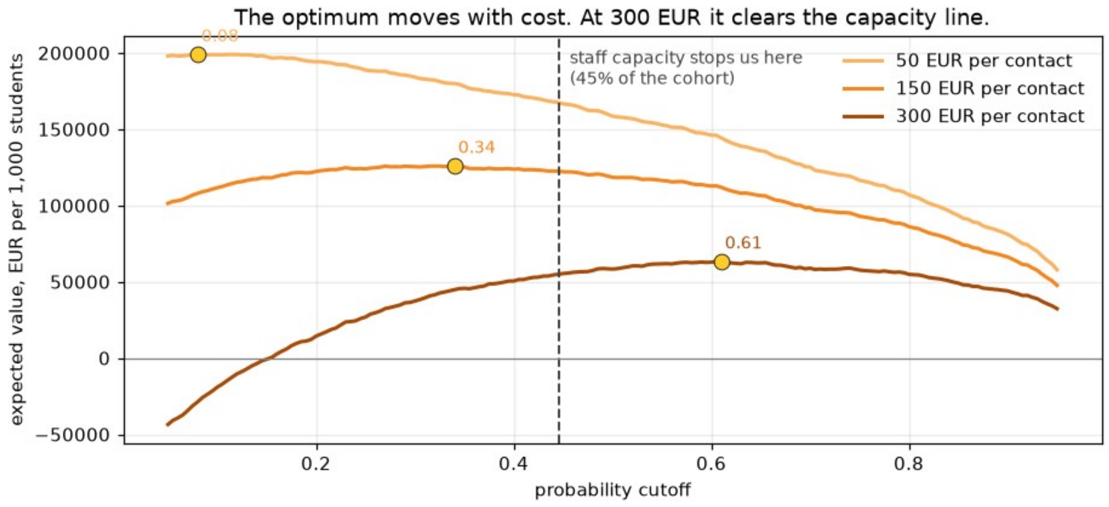
Three years of public tuition. Nothing else counted.

Reaching out works about three times in ten.

So a student the tool finds is worth roughly 630 EUR in expectation, not 2,100.

A conversation costs somewhere between 50 and 300 EUR.

The school knows this number. I do not, so all three are shown.



A conversation costs	Best case	As deployed
50 EUR	199,022 EUR	171,942 EUR
150 EUR	**125,289 EUR**	**123,595 EUR**
300 EUR	58,967 EUR	51,074 EUR

Value per 1,000 students. Best case is the most profitable cutoff. As deployed is the 1 in 2 the team can staff. The gap at 150 EUR is 1,700 EUR, so fitting the tool to the team is nearly free.

At 150 EUR a conversation, a save returns four times what a contact costs. The money says reach out broadly. There is barely any triage left to do.

The price, and who pays it

172 students in every 1,000 are contacted who were never going to leave.

That is the real cost of this tool, and it is not paid in euros. It is paid by a student who is pulled into a retention conversation they did not need, and who has just been told, in effect, that a system thinks they are failing.

The number can be driven down. Flag fewer students and fewer graduates get bothered. But every student cut from the list is also a leaver the team never reaches, and the 80 who are missed today becomes 120.

not flagged	flagged
437 correctly left alone	**172** contacted for nothing
80 missed	**311** caught

This is the trade, and it does not go away.

There is no setting where the tool finds more leavers and bothers fewer graduates. Moving one moves the other. The only question is where the school wants to sit, and that is a judgement about students, not a calculation.

Which is why the list goes to a person, and the person decides who to call.

The tool is worse at its job for women

The tool was checked across gender, age, scholarship, and debt. Three of the four came back clean, in the sense that the tool reports gaps that already exist in the world rather than inventing them. Gender did not.

Share of the students who go on to leave that the tool actually finds.

3 in 10

women who go on to drop out are never on the list at all. For men it is closer to one in eight.

This does not show up in any count of who gets flagged.

The tool flags men twice as often as women, so a check on flagging rates says men are over-contacted and stops there. That check is not wrong. It is just answering a different question.

The harm only appears when you stop counting who gets contacted and start counting who gets missed.

The students this tool exists to reach are being missed, and they are being missed unevenly.

Closing that gap costs about 30 students

There is a way to correct it. It works. It is not free, and the school should see the bill before it decides.

Leavers found, per 1,000 students.

Thirty fewer students are reached, every thousand.

That is a real cost and it should be written down plainly rather than buried in an appendix. Thirty students who would have been caught, now will not be.

What it buys is that those thirty are not disproportionately women.

The correction does not make the tool better. It makes it evenly bad.

The option the school should not take

A second correction closes the same gap and looks better on paper, because overall accuracy goes up.

It gets there by contacting fewer students of every kind. It closes the gap by helping fewer people, and it costs seventy fewer leavers found instead of thirty.

Fixing a gap by helping fewer people is not a fix.

This is the school's decision, not the model's.

Thirty students is the price of the tool being as good at finding women as it is at finding men. Nobody but the school can say whether that is worth paying.

The recommendation is to pay it. But the number is on the table, not hidden.

How to put it in, without betting the school on it

Nothing here asks anyone to trust the tool on day one. Each phase earns the next.

ONE TERM

1. Watch it, do not act on it

The tool produces its list. Nobody is contacted. At the end of term, compare who it flagged against who actually left, and against who the tutors would have worried about anyway.

If it does not beat the tutors, it does not deploy.

ONE YEAR

2. One department, real contact

A single department runs it for real. The list goes to a tutor, the tutor decides who to call.

This measures the thing nobody has measured yet, which is whether reaching out actually retains anyone. Every number on slide 5 rests on an assumption of three in ten. This is where that gets checked.

ONGOING

3. Full rollout, with a standing check

School-wide, with the gender gap re-measured at every intake and reported to whoever owns this decision.

The gap was measured on one cohort of 130 women. It is not the kind of thing that stays fixed on its own.

The tool is cheap to run and expensive to trust. So buy the trust in stages.

Three conditions, and the ask

1. **A student never sees their score.**

   Not on a portal, not in a letter, not read out by a tutor. A person told they are seventy percent likely to fail has been handed a prediction, not help.

2. **Fix what the school can actually change.**

   The second strongest signal in the whole tool is whether the fees are paid. A payment plan is cheaper than a tutor and it addresses the thing itself, not the symptom. If the tool is going to point at unpaid fees, the school should do something about unpaid fees.

3. **Re-measure the gap every intake.**

   The gender gap was measured once, on one cohort. Put it in the annual reporting line and give it an owner.

THE ASK

Approve phase one.

One term. The tool runs, nobody is contacted, and at the end we know whether it beats a tutor's judgement.

It costs a term of nobody's time and it buys the only thing worth buying at this stage, which is evidence.

And approve the shape of the thing.

A ranked list for the tutoring team. Read by a person, acted on by a person. Not a label attached to a student's record, which is a thing that follows them around.

The tool works. It should not be trusted blindly, and it does not need to be. It only has to point a tutor at the right student, early enough to matter.